Explicit symplectic reverser and overgroup bounds

Human Compression project

This mathematical component is for $G = \mathrm{PSp}_{2m}(q)$, $m \geq 2$. The complete Singer maximal-overgroup classification is retained from Bubboloni–Spiga–Weigel Lemma5.1, citing Malle–Saxl–Weigel Theorem1.1. Its full primary statement has been acquired and read. It includes a unitary overgroup omitted from the target's Table4 and explicitly covers the low ranks; its two extra parameters are among the acquired finite exceptions. The earlier Malle–Saxl–Weigel proof itself was not obtained or reread. All source finite exceptions are acquired from the finite35 certificates. The lower bounds are acquired from the accepted classical lower theorem, and the nonalternating isometry group calculation from the companion linear/unitary proof. This proves the symplectic positive branch relative to that explicit classification premise, not the complete original disposition. Write $\nu_2(H) = \mathrm{i}_2(H) + 1$, the number of solutions of $h^2 = 1$.

1 The torus and all possible reversers

Put $E = \mathbb{F}_{q^{2m}}$, $E_0 = \mathbb{F}_{q^m}$, and $\bar{u} = u^{q^m}$. Choose $\gamma \neq 0$ with $\bar{\gamma} = -\gamma$, and take the alternating form

$$B(u, v) = \mathrm{Tr}_{E/\mathbb{F}_q}(\gamma u \bar{v}).$$

Multiplication by a generator ω of the norm-one group of E/E_0 is an isometry X of order $q^m + 1$. Its projective image x has order $M = (q^m + 1)/\gcd(2, q - 1)$. In $J = \mathrm{PGSp}_{2m}(q)$ its torus centralizer has order $D = q^m + 1$.

Here is the regularity and outer-coset check. For $m \geq 2$ in the unexcluded parameters, $M > q^{m/2} + 1$. A projective centralizer normalizes the field algebra generated by X and induces a q^j-power map on E. It follows that $M \mid q^j - 1$. For $0 < j < 2m$, the gcd with $q^m + 1$ is either at most two or is $q^{\gcd(m,j)} + 1$, where the gcd is a proper divisor of m. It is at most $q^{m/2} + 1$, a contradiction. Thus the centralizer consists of multiplication maps, and its projective similitude part is the torus of order D. The same argument applied after squaring shows that a natural projective inverter induces exactly the q^m-power map. It therefore has a lift $h : u \mapsto c\bar{u}$. This is a similitude precisely when $a = c\bar{c} \in \mathbb{F}_q^*$; its multiplier is $-a$ and its square is aI.

For $m \geq 3$, the quotient $\mathrm{Aut}(G)/J$ is the cyclic field group. A nontrivial involutory field projection has f even, $q = p^f$, and exponent $f/2$. Eigenvalue comparison would imply $M \mid p^{f/2+fj} + 1$ for some j, and hence $M \mid q^{2j+1} - 1$. But $\gcd(q^m + 1, q^{2j+1} - 1) \leq 2$, impossible. The same excludes the ordinary field involution for $m = 2$. Thus all involutory reversers are natural, except possibly the exceptional graph-field coset of $\mathrm{Sp}_4(q)$ in characteristic two. There are at most D reversers in any one coset, since their pairwise quotients centralize x.

For $\mathrm{Sp}_4(q)$ in characteristic two, the outer group is cyclic of order $2f$. If f is even its involution is a field automorphism and is excluded above. If f is odd, the additional involutory coset has fixed group $\mathrm{Sz}(q)$, by Aschbacher–Seitz Section19, (19.3)–(19.5). These exact outer-automorphism statements, with the 1978 erratum retained, are borrowed structural inputs. We may uniformly count at most $2D$ reversers in degree four and characteristic two; all other cases use D.

2 Their fixed groups and involutions

Suppose first that q is odd. Since $h^2 = aI$ and its multiplier is $-a$, h is skew-adjoint for B. If a is a square, its two m-dimensional eigenspaces are paired totally isotropic spaces, and the isometric centralizer is $\mathrm{GL}_m(q)$. If a is nonsquare, the algebra $\mathbb{F}_q[h]$ is $\mathbb{F}_{q^2}$ and the same pairing gives a hermitian form; the centralizer is $\mathrm{GU}_m(q)$. Projective commutation permits a lift to send h to $-h$, so $C_G(\alpha)$ is contained in the corresponding normalizer, modulo $\langle -I \rangle$, with quotient of order at most two over the linear or unitary part. For $m \geq 3$, its image on the derived projective group is a subgroup of $\mathrm{Aut}(\mathrm{PSL}_m(q))$ or $\mathrm{Aut}(\mathrm{PSU}_m(q))$. The kernel is cyclic of order at most $q + 1$. The extra coset acts as inverse transpose, a genuine outer graph map in these degrees; it introduces no further kernel. Lawther–Liebeck–Seitz Proposition1.3 consequently gives

$$\mathrm{i}_2(C_G(\alpha)) \leq (q + 1)\{2(q^{a_m} + q^{a_m - 1}) + 1\} - 1, \qquad a_m = (m^2 + m - 2)/2. \tag{1}$$

The entire kernel size is charged: it is not assumed central in the normalizing coset.

For q even, rescale a natural inverter to have square one. Hilbert90 for E/E_0 gives the required conjugating multiplier. Multiplying that multiplier by an element of E_0 makes its norm one, since squaring is onto in E_0; thus the conjugation is by a torus isometry and takes the inverter to $u \mapsto \bar{u}$. Its square-zero part has rank m. The induced form is nonalternating, since $B(u, \bar{u}) = \mathrm{Tr}_{E/\mathbb{F}_q}(\gamma u^2)$ is not identically zero. In a symplectic basis it is therefore

$$t = \begin{pmatrix} I & I \\ 0 & I \end{pmatrix}.$$

Every commuting isometry has diagonal blocks A, A, where $AA^t = I$, and its off-diagonal block can be written DA^{-t} for a symmetric matrix D. Equivalently the centralizer is

$$\mathrm{Sym}_m(q) \rtimes J_m(q), \qquad A : D \longmapsto ADA^t,$$

where J_m is the isometry group of a nonalternating symmetric form. This follows directly from the two block commutation equations and the symplectic pairing equation; the kernel is the additive space of symmetric matrices, of order $q^{m(m+1)/2}$. In particular

$$\mathrm{i}_2(C_G(\alpha)) \le q^{m(m+1)/2}\nu_2(J_m(q)) - 1. \tag{2}$$

The identity image is included. The acquired elementary description gives $J_m = \mathrm{Sp}_{m-1}$ for odd m and

$$\nu_2(J_m) \le q^{m-1} + q^{m-2}\mathrm{i}_2(\mathrm{Sp}_{m-2}) \quad (m \text{ even}).$$

For $m = 2$, J_2 is the additive group of order q. Each nonidentity $A \in J_2$ fixes a two-dimensional subspace of the three-dimensional space $\mathrm{Sym}_2(q)$: conjugate its single Jordan block to $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ and solve $ADA^t = D$. Thus the exact centralizer involution count is

$$\mu_2(q) = q^3 + (q-1)q^2 - 1 = 2q^3 - q^2 - 1. \tag{3}$$

In the exceptional outer coset, the Suzuki group has one involution class, with centralizer q^2, so its involution count is $(q-1)(q^2+1) < \mu_2(q)$. Its class order is retained from the primary involution inventory. This proves the same bound for every reverser.

The odd-characteristic degree-four case. Here the graph on the derived degree-two group is inner, so use an exact count rather than the kernel assertion in (1). In the split case take

$$\mathrm{GL}_2(q) : A \mapsto \mathrm{diag}(A, A^{-t}), \qquad w = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}.$$

Then $w^2 = -I$ and w acts on A by A^{-t}. In the nonsplit case the same normalizer is described by $\mathrm{GU}_2(q)$ and a semilinear w with square $-I$, acting by conjugation on the field entries. Such a w is obtained by multiplying coordinate conjugation by a scalar of norm -1. Modulo $\langle -I \rangle$, an element of the linear part is an involution when $A^2 = I$ or $-I$. Splitting their two eigenspaces, or using the irreducible quadratic centralizer when needed, gives

	$q \equiv 1 \pmod 4$	$q \equiv 3 \pmod 4$
$\mathrm{i}_2(\mathrm{GL}_2/\{\pm I\})$	$q(q+1)+1$	q^2
$\mathrm{i}_2(\mathrm{GU}_2/\{\pm I\})$	q^2	$q(q-1)+1.$

In the other coset, square $\pm I$ means $A = \pm A^t$. There are $q^2(q-1)$ nonsingular symmetric and $q-1$ nonsingular alternating linear matrices, hence $(q-1)(q^2+1)/2$ projective solutions. For unitary matrices, symmetric ones correspond by finite-field descent to the two orthogonal forms, so their number is $|\mathrm{GU}_2|(1/|O_2^+| + 1/|O_2^-|) = q^2(q+1)$; there are $q+1$ alternating ones. This gives $(q+1)(q^2+1)/2$ solutions. Consequently, for $q \ge 5$,

$$\mathrm{i}_2(C_G(\alpha)) \le \tfrac{1}{2}(q+1)(q^2+1) + q^2 + q + 1 < q^3. \tag{4}$$

The inequality reduces to $q^3 - 3q^2 - 3q - 3 > 0$, already positive at five and increasing thereafter.

3 Retained covering and the uniform inequalities

Retain the complete list of maximal overgroup *types* from Bubboloni–Spiga–Weigel Lemma5.1 for the full Singer lift X in $\mathrm{Sp}_{2m}(q)$, $m \geq 2$, $(m,q) \neq (2,2)$: the extension-field types $\mathrm{Sp}_{2m/r}(q^r).r$ for primes $r \mid m$; additionally $O^-_{2m}(q)$ when q is even; and additionally $\mathrm{GU}_m(q).2$ when m, q are odd. Here $\mathrm{GU}_m(q)$ is the isometry group on the m-space over $\mathbb{F}_{q^2}$. The only extra cases in that statement are $(m,q) = (4,2)$ with a $\mathrm{PSL}_2(17)$ overgroup and $(m,q) = (2,3)$ with a $2^{1+4}_-.A_5$ overgroup. Both are paid by the acquired finite35 certificates, so no uniform inequality is applied there. The unitary type is absent from the target Table4, so that printed list is not used as a complete cover. Passing to projective groups causes no new case: the full inverse image of a subgroup containing x contains X and the scalar centre.

Here are the multiplicities for the fixed element, rather than just the conjugacy types. In an extension-field normalizer, X^r is linear over $K = \mathbb{F}_{q^r}$. Its order is $(q^m+1)/\gcd(r, q^m+1)$ and is greater than $q^{m/2}+1$ in all unexcluded parameters. For $m \geq 6$ use $q^{m/2} - 1 > m$; for $m = 2, 3, 4$ direct substitution at the smallest retained q proves the inequality. For $m = 5$, the only remaining smallest case is $q = 2$, where $r = 5$ does not divide 33. The same eigenvalue gcd argument as above proves that X^r is irreducible. Thus $K \subseteq C_{\mathrm{End}_{\mathbb{F}_q}(E)}(X^r) = E$, where a subfield of this order is unique. The full normalizer of this one field structure is one subgroup, so there is at most one extension-field term for each r. For the unitary type, apply the same argument to X^2. Its unique quadratic field is the subfield $\mathbb{F}_{q^2} \subset E$; since m is odd, the bar map restricts to its nontrivial involution. The alternating trace form determines the hermitian form, and the full normalizer of this field structure is the one unitary normalizer. In characteristic two, two X-invariant quadratic forms polarizing to B differ by the square of a linear functional fixed by X. Irreducibility excludes a nonzero such functional. Hence the orthogonal overgroup is unique as well. The normalizer of $\langle X \rangle$ preserves each of the unique prime-degree extension fields, so it is included even when m is prime. No uncharged normalizer term is discarded.

For the extension-field term, the actual scalar quotient embeds in the automorphism group of $\mathrm{PSp}_{2m/r}(q^r)$. For odd r, its involutions lie in the linear part, but the same upper bound suffices. The bound is

$$E_r < 2\{q^{m^2/r+m} + q^{m^2/r+m-r}\}.$$

The additional even-characteristic orthogonal term is bounded by $2(q^{m^2} + q^{m^2-1})$; its isometry group embeds in $\mathrm{Aut}(\Omega^-_{2m}(q))$. This includes $m = 3$ through $D_3 = A_3$ and $m = 2$ through $O^-_4(q) = \mathrm{SL}_2(q^2).2$. For the additional odd-characteristic unitary normalizer, the projective image acts on $\mathrm{PSU}_m(q)$ with cyclic kernel of order at most $q+1$; its extra coset is a graph automorphism for $m \geq 3$. Thus its entire involution contribution is at most the right side of (1). All smaller groups used here are simple. Thus, with at most m prime terms, their total is at most $3mq^{m^2/2+m}$; for $m = 3$ it is at most $3q^6$. In characteristic two add $3q^{m^2}$ for the orthogonal term.

For odd q and $m \geq 3$, the reversing contribution from (1) together with that unitary term when present is less than $5q^{(m^2+3m)/2}$. Indeed, charge $(q^m+2)\mu$ uniformly, and use $q^m + 2 \leq (29/27)q^m$, $q+1 \leq 4q/3$, and $2(q^{a_m} + q^{a_m-1}) + 1 \leq 3q^{a_m}$. Relative to the acquired lower scale q^{m^2+m}, at $m = 3$ the total is at most $5/q^3 + 3/q^6 < 1/4$ for $q \geq 3$. For $m \geq 4$ it is at most

$$5\,3^{-m(m-1)/2} + 3m\,3^{-m^2/2} < 1/4.$$

Both summands decrease from $m = 4$, where their sum is less than 0.009. The accepted quarter lower bound proves the strict counting inequality.

For q even the accepted explicit class lower bound is stronger: $\mathrm{i}_2(G) > (2/3)q^{m^2+m}$. For odd $m \geq 5$, (2) and LLS give reversing contribution less than $5q^{(3m^2+6m-1)/4}$. Its ratio, with the other terms, is at most

$$5\,2^{-(m-1)^2/4} + 3\,2^{-m} + 3m\,2^{-m^2/2} < 2/3.$$

At $m = 5$ the sum is less than 0.41 and all terms decrease. For even $m \geq 6$, the bound $\nu_2(J_m) < 4q^{m^2/4+m/2-2}$ follows from its displayed formula and LLS. The ratio is instead at most

$$5\,2^{-(m^2/4-m+2)} + 3\,2^{-m} + 3m\,2^{-m^2/2} < 2/3;$$

at $m = 6$ it is less than 0.204, again decreasing. The numerical LLS bound for the nonsimple $\mathrm{Sp}_4(2)$ is paid by its exact 75 involutions; no simplicity premise is used there.

For $m = 3$ in characteristic two, $\nu_2(J_3) = q^2$, so the reversing term is $(q^3+1)(q^8-1)$. At $q \geq 4$ the total ratio is bounded by $(1+q^{-3})/q + 2(1+q^{-1})/q^3 + 3/q^6 < 2/3$. For $m = 4$, the exact $\nu_2(J_4) = q^4 + q^3 - q^2$ bounds the

reversing term by $5q^{18}$. At $q \geq 4$ the total ratio is at most $5/q^2 + 3/q^4 + 12/q^8 < 2/3$. The unused $q = 2$ cases are finite35 inputs.

For $m = 2$, odd $q \geq 5$, the accepted class lower bound exceeds $q^5(q-1)/2$. Equations (4) and the single extension-field bound give total less than $q^5 + 2q^4 + q^3 + 2q^2$, which is smaller already at $q = 5$; after division by q^5, the left side $(q-1)/2$ increases and the right side decreases. For even $q \geq 8$, use the exact class lower bound $(q^2-1)(q^4-1)$, the at most $2D$ reversers, and two overgroup bounds. Their difference is at least

$$q^6 - 4q^5 - 3q^4 - 4q^3 - q^2 + 3 > 0.$$

Divide by q^6 and drop its positive last term; the resulting lower bound is positive at eight and increasing. The other degree-four parameters are in finite35, with the nonsimple $\mathrm{Sp}_4(2)$ excluded from the original positive simple-group claim.

An involution outside the covered proper subgroups generates G with x. Any reversing automorphism fixing this partner squares to the identity, so all such obstructions have been included. The strict inequalities therefore prove the symplectic existence assertion relative to the explicit retained covering classification, without the source's unexpanded arithmetic or its uniform single-inverter-class assertion.

Attribution and limits. The target is Leemans–Liebeck, arXiv1603.07713v3, Section4. The missing unitary overgroup was identified by root from BLW Lemma3.1 before this component was frozen; this author acquired the full all-rank BSW Lemma5.1 to pay the remaining low-rank interface. Earlier drafts and their arithmetic are retained. The projective coset organization was suggested by root; the present symplectic trace-form, field exclusion, exact degree-four fibres and inequalities are this author's new derivations. AS1976 Section19 and its 1978 erratum supply the exceptional outer fixed groups; BT Proposition2.3/Table5 supplies the Suzuki class count; LLS2002 Proposition1.3 is the numerical upper theorem. The finite35 and classical lower components retain their separate authorship and distinct reviews. This is neither a proof of maximal subgroup classification nor a whole-source admission.

References

[1] D. Leemans and M. W. Liebeck, *Chiral polyhedra and finite simple groups*, arXiv:1603.07713v3, Section4, Tables3–5.

[2] R. Lawther, M. W. Liebeck and G. M. Seitz, *Fixed point ratios in actions of finite exceptional groups of Lie type*, Pacific J. Math. 205 (2002), Proposition1.3.

[3] D. Bubboloni, M. S. Lucido and Th. Weigel, *2-Coverings of classical groups*, arXiv:1102.0660, Lemma3.1; there citing G. Malle, J. Saxl and T. Weigel, *Generation of classical groups*, Geom. Dedicata 49 (1994), 85–116, Theorem1.1.

[4] D. Bubboloni, P. Spiga and Th. Weigel, *Normal* 2*-coverings of the finite simple groups and their generalizations*, arXiv:2208.08756v1, Lemma5.1, physical page40. The stated classification is retained; its earlier MSW proof was not obtained.

[5] M. Aschbacher and G. M. Seitz, *Involutions in Chevalley groups over fields of even order*, Nagoya Math. J. 63 (1976), 1–91, Section19; with the authors' 1978 corrections.

[6] T. C. Burness and A. R. Thomas, *On the involution fixity of exceptional groups of Lie type*, arXiv:1707.03562v2, Proposition2.3/Table5.